Project Management Plan

Tiny code (SQL) - Fine-Tuning Lightweight Language Models for BigQuery SQL Generation

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Project Overview (AI590)

- **Project Title**: Tiny code (SQL) - Fine-Tuning Lightweight Language Models for BigQuery SQL Generation
- **Objective**: Create a high-quality, augmented 10k-sample dataset ready for text-to-SQL model fine-tuning (e.g., using Gemma or similar), demonstrate augmentation value via EDA, and deliver a complete ML capstone report + presentation.
- **Team**: Group 8 : Team Member1 , Team Member2 , Team Member3
- **Framework**: CRISP-DM (Business Understanding → Data Understanding → Data Preparation → Modeling → Evaluation → Deployment), adapted to fit the 7-week milestones.
- **Key Deliverables** (by end of Week 7):
  - GitHub repo (code, notebooks, processed dataset, readme),
  - Project report (with drafted sections compiled),
  - Recorded presentation (10–15 min video),
  - Peer evaluation inputs.
- **Tools**: Colab/Jupyter, GitHub, Google Drive, sqlglot, datasets, pandas, matplotlib/seaborn.

Aligned 7-Week Plan with Milestones, CRISP-DM Mapping, Tasks, Assignments, and Reviews

Each week ends with the required module deliverable. Include **peer code/writing reviews** (30–60 min + comments in Colab/GitHub) for major outputs. Presentation recording is scheduled in Week 7.

- **Week 1: Module 1 – Form Teams, Select Project Topic and Dataset, Introduction Draft Completed CRISP-DM**: Business Understanding
  - Confirm team roles and communication (e.g., Slack/Discord).
  - Finalize topic: Synthetic data augmentation for text-to-SQL in BigQuery dialect.
  - Load/inspect Gretel dataset on Hugging Face; select 10k train slice.
  - Draft Introduction section: problem statement (challenges in text-to-SQL, need for BigQuery-safe data), project goals, significance for applied AI/LLM fine-tuning.
  - **Assigned**:
    - Team Member1: Dataset selection + initial load verification.
    - Team Member2: Introduction draft writing.
    - Team Member3: Scope/goals + references.
  - **Review**: Team review of Introduction draft (end Week 1).
  - **Output**: Introduction draft (1–2 pages), shared repo setup.

- **Week 2: Module 2 – Project Management Plan CompletedCRISP-DM**: Business Understanding (project plan) + initial Data Understanding
  - Finalize this detailed plan (assignments, timeline, risks).
  - Set up shared GitHub repo + Colab structure.
  - Preliminary data overview: load 10k subset, basic stats (shape, columns, nulls).
  - **Assigned**: All 3 (collaborative); Team Member1 leads repo setup.
  - **Review**: Self-review of plan + initial data notebook.
  - **Output**: Completed project management plan document (this or similar), repo initialized.

- **Week 3: Module 3 – Perform Data Cleaning and Exploratory Data Analysis, Dataset Summary Draft CompletedCRISP-DM**: Data Understanding + initial Data Preparation

  - Data cleaning: handle any parsing issues in sql/prompt/explanation/schema.
  - Full EDA on raw data: sql_complexity pie, task_type bar, keyword frequency, prompt/SQL length hists (as in your code).
  - Save plots + insights.
  - Draft Dataset Summary: description, stats, quality issues, visualizations. **Assigned**:
    - Team Member2: Lead EDA + visualizations.
    - Team Member1: Cleaning + stats.
    - Team Member3: Summary draft writing + interpretation.
  - **Review**: Peer code review of EDA notebook + writing feedback on summary (end Week 3).
  - **Output**: EDA notebook, saved PNGs, Dataset Summary draft.

- **Week 4: Module 4 – Select Machine Learning Methods and Tools, Background Section Draft CompletedCRISP-DM**: Data Preparation + Modeling (planning)
  - Select methods/tools: sqlglot for transpilation, random augmentation layers, Hugging Face datasets, pandas/matplotlib for EDA; plan for future fine-tuning (e.g., LoRA on Gemma).
  - Implement core augmentation function (apply_tinycode_logic): transpilation + SAFE_CAST, distractors (20%), grounding hints, style prefixes, Gemma template.
  - Apply to create tinycode_train_ds; save processed dataset.
  - Draft Background section: literature on text-to-SQL, synthetic data, BigQuery best practices, augmentation techniques.
  - **Assigned**:
    - Team Member1: Transpilation/normalization + tool selection.
    - Team Member2: Schema pruning + grounding + Background research/draft.

        - Team Member3: Style aug + template + integration.
    - **Review**: Full team code review of augmentation function + sample outputs; writing review of Background (end Week 4).
    - **Output**: Processed dataset, augmentation notebook, Background draft.

- **Week 5: Module 5 – Begin Model Training/ML System Work, Methods Section Draft CompletedCRISP-DM**: Modeling + initial Evaluation
    - Treat augmentation pipeline + EDA as core "ML system" (data prep for modeling).
    - Begin "training-like" work: run full pipeline, generate before/after EDA comparisons (e.g., length/distribution shifts, keyword changes post-augmentation).
    - Draft Methods section: detailed description of data prep, augmentation layers, EDA methods, tools.
    - **Assigned**:
        - Team Member2: Lead before/after comparisons + Methods draft.
        - Team Member1 & Team Member3: Support pipeline runs + section contributions.
    - **Review**: Peer review of updated EDA + Methods draft (end Week 5).
    - **Output**: Augmented EDA notebook, Methods draft.

- **Week 6: Module 6 – Complete Model Training/ML System Work, Results Section StartedCRISP-DM**: Modeling (finalize) + Evaluation
    - Finalize pipeline: error handling, logging, verification of 10k samples.
    - Complete Results: quantitative (e.g., % transpilation success, diversity metrics), qualitative samples, all plots (before/after).
    - Start Results section draft: present findings, insights on augmentation effectiveness.
    - **Assigned**:
        - Team Member3: Lead Results drafting + final pipeline polish.
        - Team Member1: Verification + metrics.

        - Team Member2: Plots + interpretation.
    - **Review**: Team review of Results content + code (end Week 6).
    - **Output**: Final pipeline, Results section started.

- **Week 7: Module 7 – Project Presentation, GitHub Repository, Report, and Peer Evaluation CompletedCRISP-DM**: Evaluation + Deployment
    - Compile full report: combine Introduction + Background + Methods + Results + add Discussion/Conclusion/Future Work (e.g., next steps for actual fine-tuning).
    - Prepare slides: overview, problem, methods/augmentation, EDA results, conclusions.
    - Record presentation (all members participate; use Zoom/Loom).
    - Finalize GitHub: clean notebooks, README (setup instructions, dataset link), requirements.txt, processed data sample (or link).
    - Complete peer evaluations.
    - **Assigned**:
        - Team Member3: Lead slides + recording.
        - Team Member1: Report compilation + intro/conclusion recording.
        - Team Member2: Results slides + edit video.
    - **Review**: Dry-run presentation + final proofread (early Week 7).
    - **Output**: Final report (PDF), recorded video, polished repo, peer eval submissions.

Additional Notes

- **Effort Balance**: Weeks 3–5 are heaviest (cleaning/augmentation/EDA), typical for data-heavy ML projects.
- **Risks**: If augmentation errors delay → prioritize core transpilation + template first. Use parallel tasks (e.g., one member starts Background while others code).
- **Quality Assurance**: Mandatory peer reviews each week to catch issues early.
- **Success**: High-quality augmented dataset, insightful before/after analysis showing BigQuery readiness, cohesive report, professional presentation.

Team Tasks by Week / Milestone Diagram (Adapted TDSP-Style Matrix)

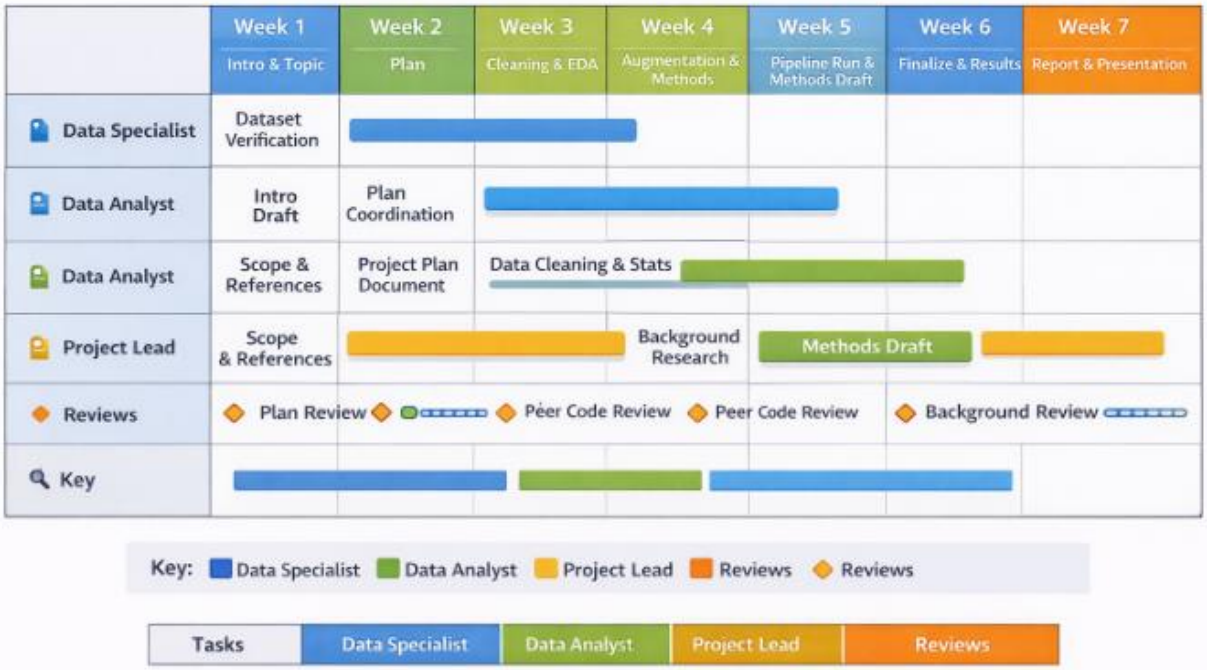
Role / Team Member	**Week 1 Module 1 Intro + Topic**	**Week 2 Module 2 Plan**	**Week 3 Module 3 Cleaning + EDA + Summary**	**Week 4 Module 4 Augmentation + Methods + Background**	**Week 5 Module 5 Pipeline Run + Methods Draft**	**Week 6 Module 6 Finalize Pipeline + Results Start**	**Week 7 Module 7 Report + Presentation + Repo**
Team Member1	Dataset load/verify Intro contributions Repo setup lead	Plan co-creation Repo structure	Cleaning + basic stats	Transpilation (sqlglot + SAFE_CAST) Tool selection	Pipeline runs + support Methods contributions	Verification + success metrics	Report compilation Intro/Conclusion recording Final repo polish
Team Member2	Intro draft lead	Plan co-creation	Lead EDA + visualizations (pie, bar, hist, keywords)	Schema pruning + grounding hints Background research/draft	Lead before/after comparisons Methods draft lead	Plots + interpretation	Results slides + video editing
Team Member3	Scope/goals + refs	Plan co-creation	Dataset Summary draft + insights	Style aug + Gemma template Full mapping/integration	Support + Methods contributions	Lead Results drafting Pipeline polish	Lead slides + recording coordination Discussion/Conclusion
All / Collaborative	Team kickoff + Introduction draft review	Self-review of plan	—	—	—	—	Dry-run presentation Final proofread + submit
Reviews (mandatory)	Intro draft review (all)	Plan + initial notebook review	Peer code/writing review (rotate)	Full code review (aug function + samples) Background review	EDA update + Methods review	Results + code review	Presentation dry-run + final proofread
Key Deliverable	Introduction draft	Project plan document + repo	EDA notebook + plots + Summary draft	Processed dataset + Background draft	Augmented EDA + Methods draft	Results section started + final pipeline	Full report + recorded presentation + GitHub repo + peer eval

Business Understanding
Data Understanding
Data Understanding
Modeling
Evaluation
Deployment
Deployment
CRISP-DM
Data Mining Process
Business Understanding
Data Understanding
Data Understanding
Data Preparation
Modeling
Evaluation
Deployment

Scrum
Daily Scrum
1-4 Week Sprint
Product Backlog
Sprint Planning
Increment
Business Understanding
Data Understanding
Data Preparation
Modeling
Sprint Review
Deployment

Team Tasks by Week / Milestone

(7-Week Academic Capstone)

Role / Team Member	Week 1 Intro & Topic	Week 2 Plan	Week 3 Cleaning & EDA	Week 4 Augmentation & Methods	Week 5 Pipeline Run & Methods Draft	Week 6 Finalize & Results	Week 7 Report & Presentation
Data Specialist	Dataset Verification	Plan Coordination	Data Cleaning & Stats	SQL Transpilation	Pipeline Runs	Success Metrics	Report Compilation
Data Analyst	Intro Draft		Lead EDA & Visuals	Background Research	Methods Draft	Success Metrics	Slides & Video Edit
Project Lead	Scope & References	Project Plan Document	Summary & Insights	Data Augmentation	Methods Draft	Results Drafting	Final Proofread
All / Reviews	Intro Review	Plan Review	Peer Code Review	Background Review	Methods Review	Results Review	Final Proofread
Key Deliverables	Introduction Draft	Project Plan & Repo	EDA Notebook & Summary	Augmented Dataset & Methods	Pipeline & Methods Draft	Results Section Started	Full Report & Presentation

Capstone Project Timeline (Gantt Chart)